R-3-5 BRAKE CLEARANCE MEASUREMENT/ADJUSTMENT

id051700664900

Preparation Before Servicing

1. Print out the measurement/adjustment value input sheet. (See MEASUREMENT/ADJUSTMENT VALUE INPUT SHEET.)

Note

- When performing the measurement/adjustment, input the measured and calculated values into the measurement/adjustment value input sheet.
- If the measurement/adjustment value input sheet has already been printed out for the other measurements/adjustments, use the sheet.

R-3-5 Brake Clearance Measurement

1. Assemble the drive plates and driven plates.

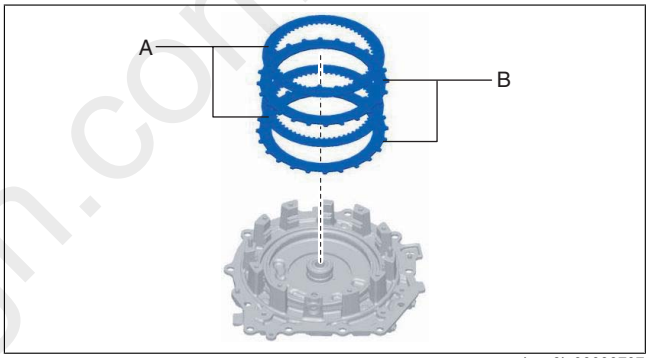
Note

- Drive plate size: Outer diameter approx. 159.8 mm {6.291 in}
- Driven plate size: Inner diameter approx. 132.2 mm {5.205 in}

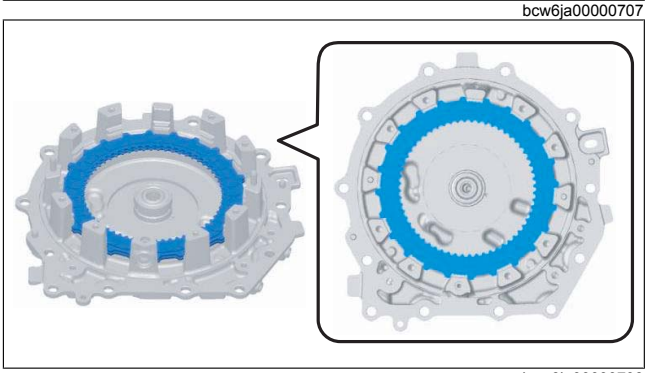
Assembly order

Driven plate—drive plate—driven plate—drive plate

A : Drive plate
B : Driven plate



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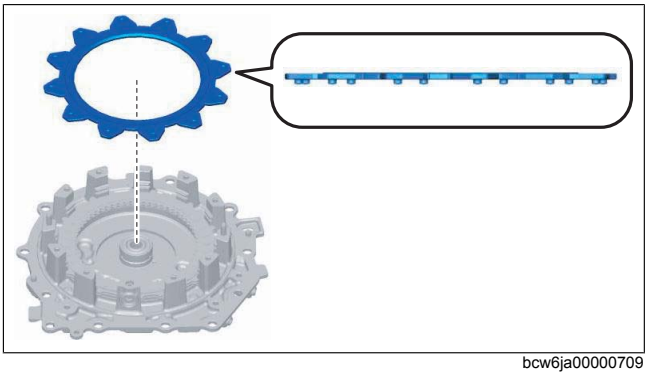


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2. Assemble the retaining plate.

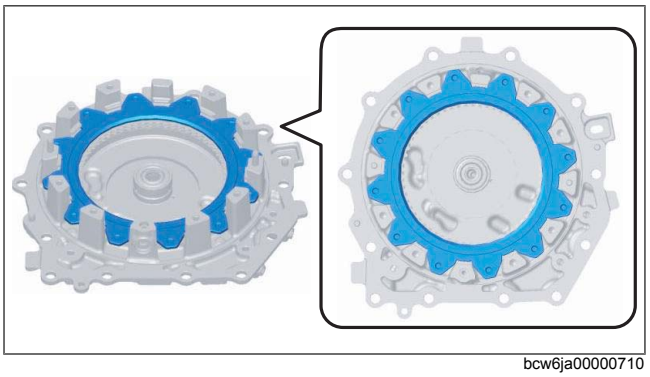
Note

- Retaining plate size: Inner diameter approx. 132.2 mm {5.205 in}



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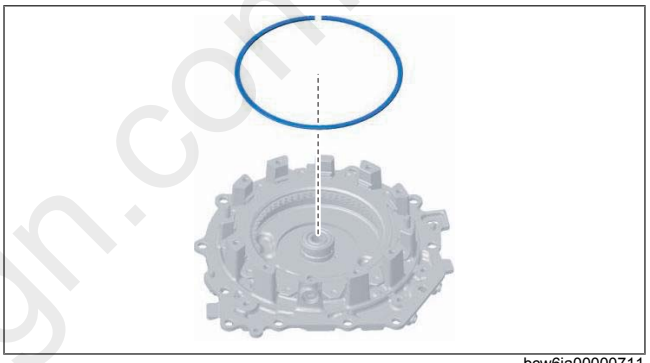


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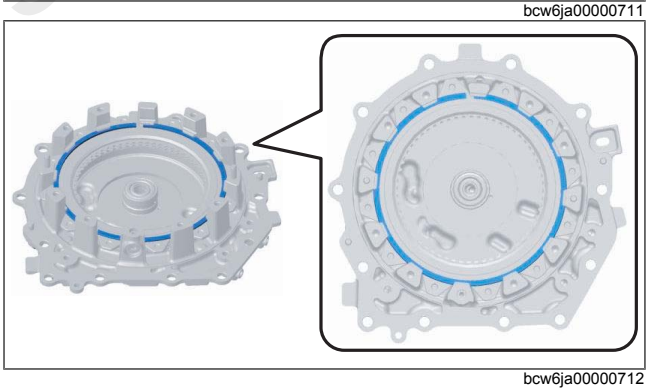
3. Assemble the snap ring.

Caution
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

Note
- Snap ring size: Outer diameter approx. 174.2 mm {6.858 in}



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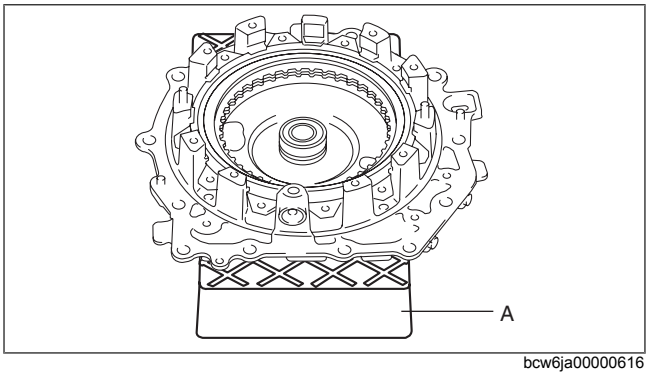


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4. Set the end cover with the assembled part on the workbench as shown in the figure.

Caution
- **To reduce measurement differences during the R-3-5 brake clearance measurement, use the rubber plates to adjust the alignment surface of the end cover with the transaxle case so that it is level.**

A : Rubber plate

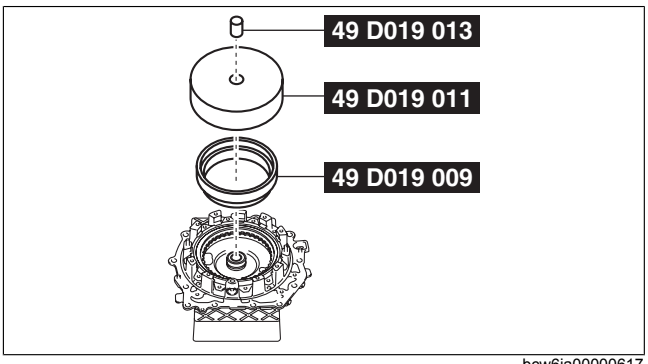


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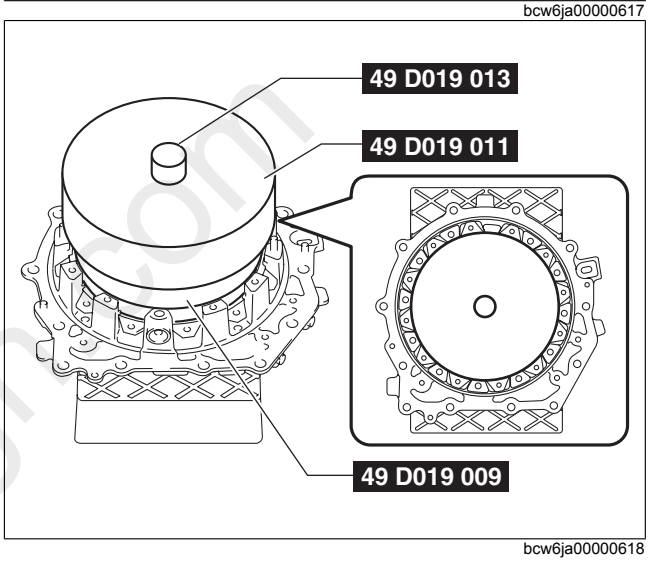
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5. Install the SSTs.

Caution
- **Install the SST (49 D019 009) so that the small diameter side is pointing towards the retaining plate.**



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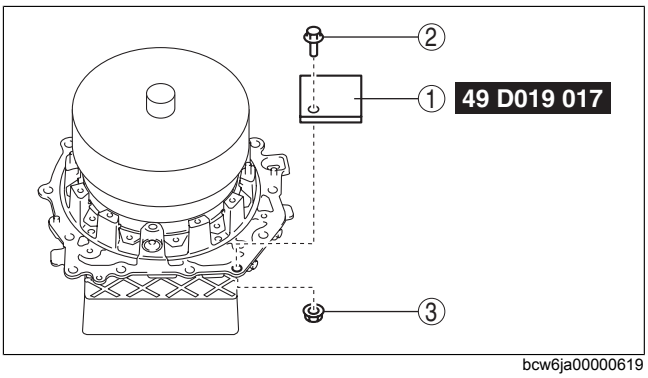


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6. Set the measuring instrument to the end cover using the following procedure.
   (1) Install the SST for securing the magnetic stand using the procedure shown in the figure.

Caution
- **If the bolt and nut are tightened with excessive force when installing the SST, the alignment surface of the end cover with the transaxle case could be damaged. Tighten the bolt and nut so that the SST does not move during the R-3-5 brake clearance measurement.**

Note
- When installing the SST to the end cover, use an M8 bolt and nut.

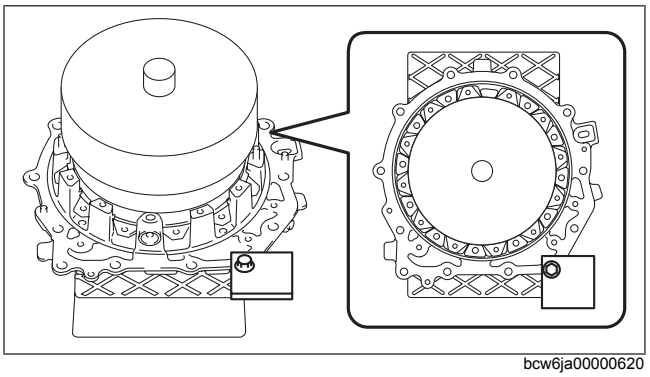


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SST installation bolt tightening torque
15 N·m {1.5 kgf·m, 11 ft·lbf} or less (tighten so that SST does not move during R-3-5 brake clearance measurement)

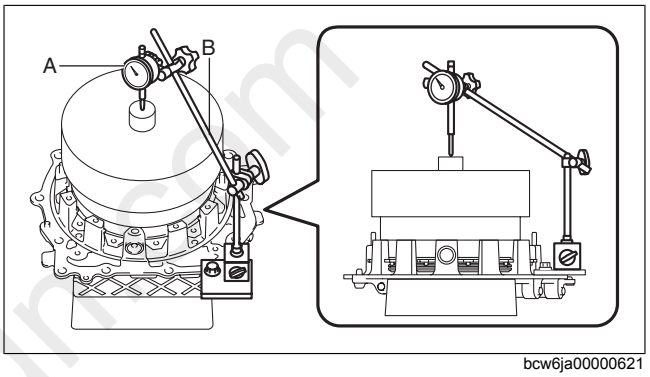
1	SST (for securing magnetic stand)
2	Bolt (M8)
3	Nut (M8)


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(2) Set the dial gauge and magnetic stand as shown in the figure.

Caution
- **To reduce measurement differences during the R-3-5 brake clearance measurement, set the dial gauge so that it is perpendicular to the alignment surface of the end cover with the transaxle case.**

A : Dial gauge
B : Magnetic stand

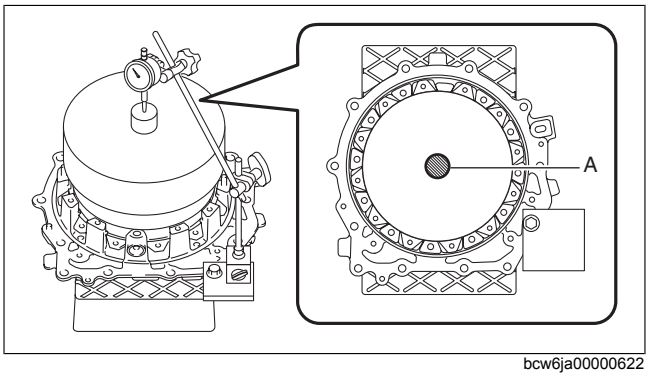


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(3) Set the dial gauge end near the center of the SST.

Caution
- **To reduce measurement differences during the R-3-5 brake clearance measurement, set the dial gauge end to within the area shown in the figure.**

A : Dial gauge end set area



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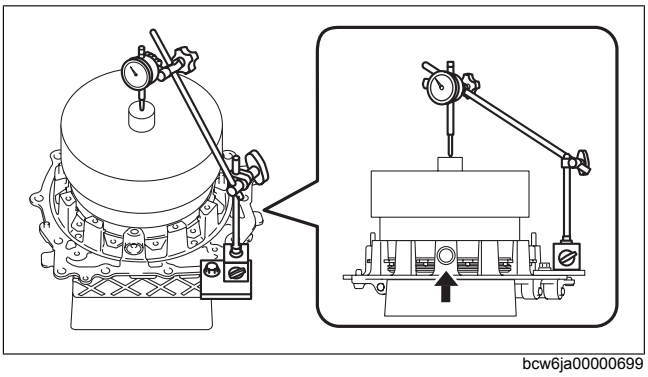
7. Measure the R-3-5 brake clearance using the following procedure:
(1) Blow compressed air into the oil passage shown in the figure to force the R-3-5 brake piston to stroke (approx. 3 times).

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**

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Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm2, 57—63 psi}



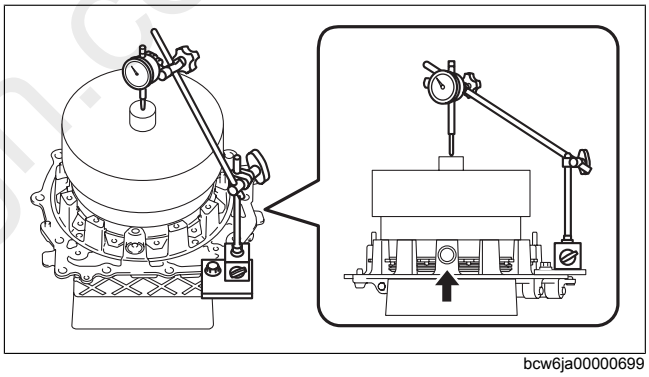
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(2) Blow compressed air into the oil passage shown in the figure to operate the R-3-5 brake piston, and read the value when the dial gauge is stabilized.

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**
- **To reduce measurement differences during R-3-5 brake clearance measurement, gently blow and release the compressed air.**

Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm2, 57—63 psi}



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(3) Input the dial gauge value, which was read while the R-3-5 brake piston was operating, into the measurement/adjustment value input sheet.
(4) Release the compressed air and read the dial gauge value while the R-3-5 brake piston is not operating.
(5) Input the dial gauge value, which was read while the R-3-5 brake piston was not operating, into the measurement/adjustment value input sheet.
(6) Perform the following calculation to calculate the R-3-5 brake clearance.
R-3-5 brake clearance = A - B
A: Dial gauge value while R-3-5 brake piston is operating
B: Dial gauge value while R-3-5 brake piston is not operating

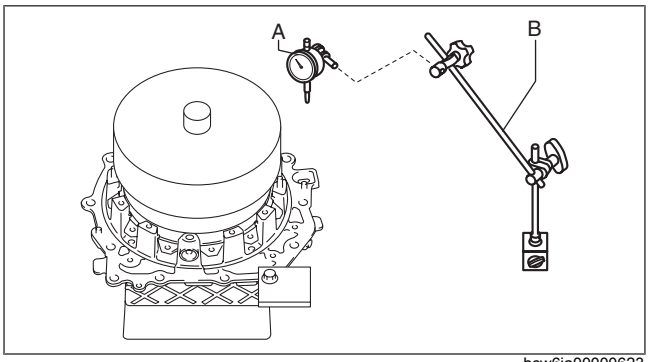
Note
Example
A: Dial gauge value while R-3-5 brake piston is operating is 1.990 mm {0.07835 in}
B: Dial gauge value while R-3-5 brake piston is not operating is 0.595 mm {0.0234 in}
R-3-5 brake clearance = 1.990 mm {0.07835 in} - 0.595 mm {0.0234 in} = 1.395 mm {0.05492 in}

(7) Input the calculated R-3-5 brake clearance into the measurement/adjustment value input sheet.
(8) Verify that the R-3-5 brake clearance satisfies the specification.

Specification
1.030—1.230 mm {0.0406—0.0484 in}

- If not within the specification, adjust the R-3-5 brake clearance. (See R-3-5 Brake Clearance Adjustment.)

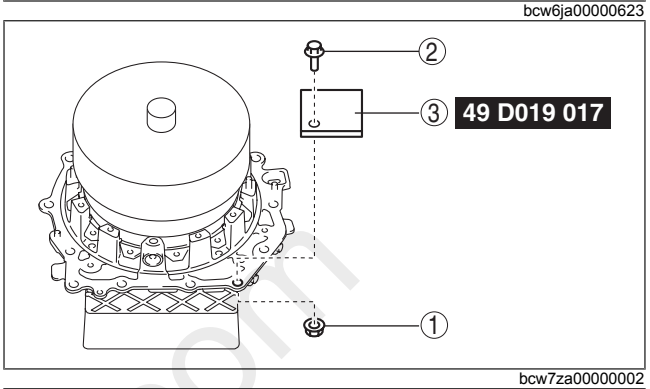
8. Remove the dial gauge and magnetic stand.
A : Dial gauge
B : Magnetic stand



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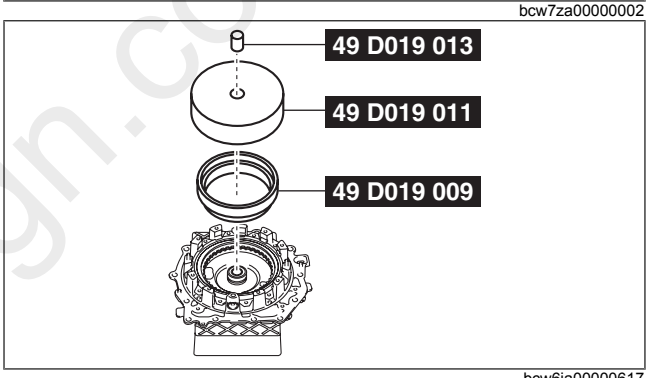
9. Remove the SST for securing the magnetic stand using the procedure shown in the figure.

1	Nut (M8)
2	Bolt (M8)
3	SST (for securing magnetic stand)



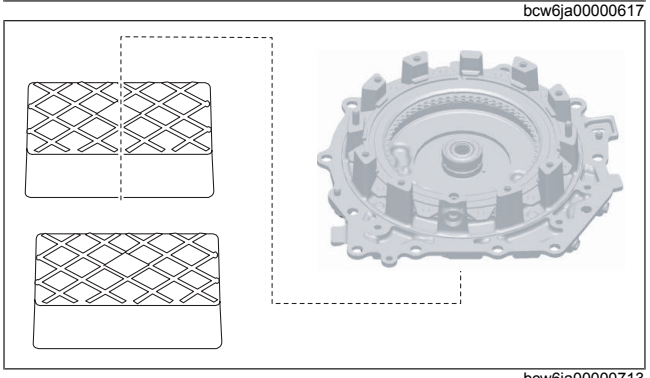
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10. Remove the SSTs.

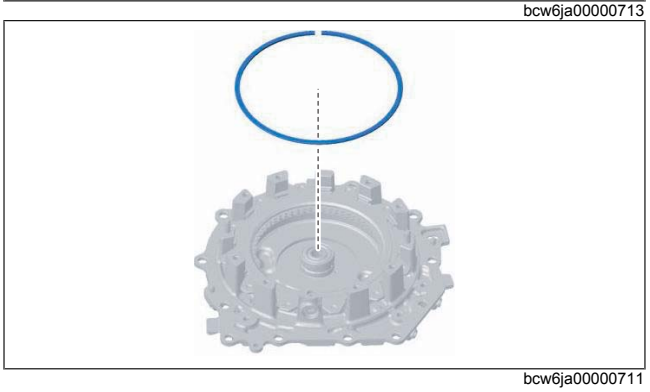


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11. Take the end cover off the rubber plates.

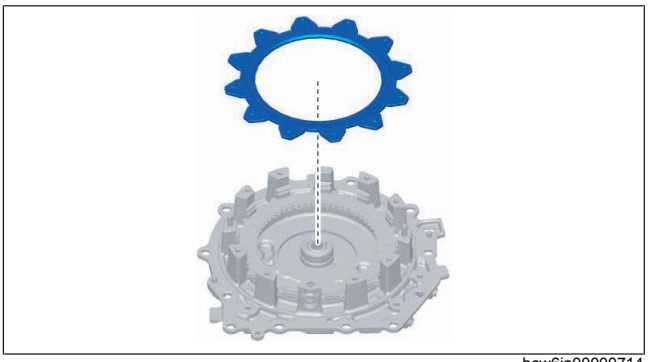

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12. Remove the snap ring.

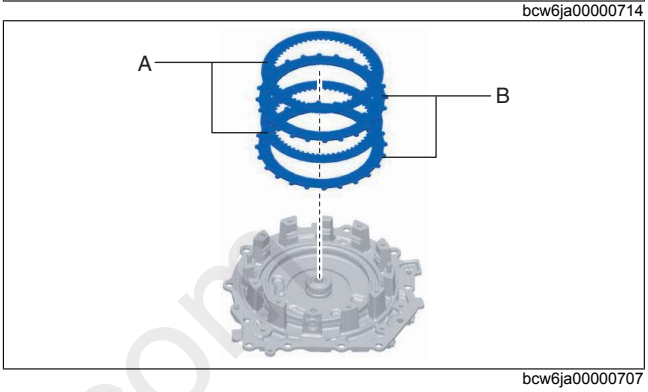

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13. Remove the retaining plate.


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14. Remove the drive plates and driven plates.
A : Drive plate
B : Driven plate

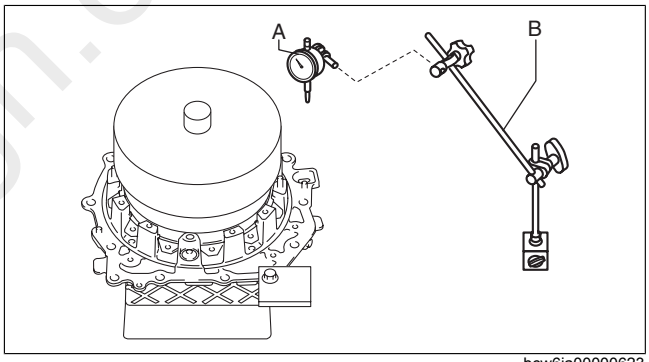


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R-3-5 Brake Clearance Adjustment

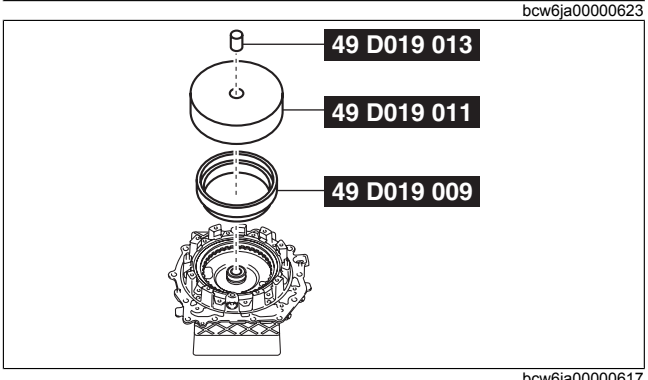
1. Remove the dial gauge and magnetic stand.

A : Dial gauge
B : Magnetic stand



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2. Remove the SSTs.

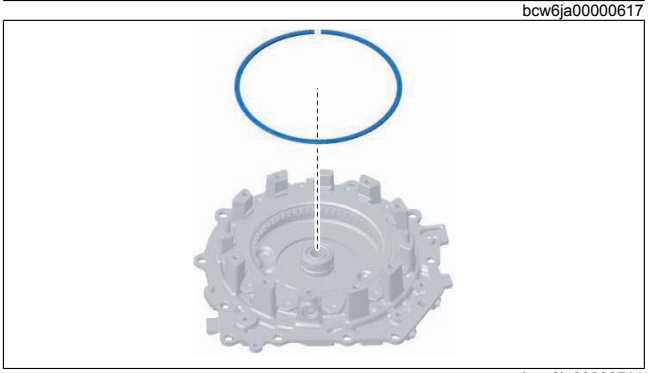


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3. Remove the snap ring.
4. Measure the thickness of the removed snap ring.

Note
- Recommended measuring instrument: Micrometer

5. Input the measured snap ring thickness into the measurement/adjustment value input sheet.
6. Perform the following calculation to calculate the R-3-5 brake clearance gap.


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Note
- The R-3-5 brake clearance gap is the difference between the R-3-5 brake clearance and the median value of the R-3-5 brake clearance specification.

R-3-5 brake clearance gap = C-G
C: R-3-5 brake clearance
G: R-3-5 brake clearance specification median value (1.130 mm {0.04449 in})

Note
Example
C: R-3-5 brake clearance is 1.395 mm {0.05492 in}
R-3-5 brake clearance gap = 1.395 mm {0.05492 in} - 1.130 mm {0.04449 in} = 0.265 mm {0.0104 in}

7. Input the calculated R-3-5 brake clearance gap into the measurement/adjustment value input sheet.
8. Perform the following calculation to calculate the optimum snap ring thickness.
Thickness of optimum snap ring = F + H
F: Thickness of removed snap ring
H: R-3-5 brake clearance gap

Note
Example
F: Thickness of removed snap ring is 2.140 mm {0.08425 in}
Thickness of optimum snap ring = 2.140 mm {0.08425 in} + 0.265 mm {0.0104 in} = 2.405 mm {0.09469 in}

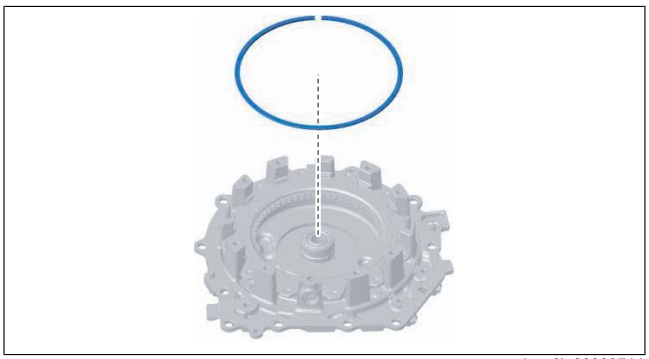
9. Input the calculated optimum snap ring thickness into the measurement/adjustment value input sheet.
10. From the following table, select a snap ring of a thickness closest to the calculated optimum snap ring thickness.

Snap ring thickness
2.5 mm {0.098 in}
2.4 mm {0.094 in}
2.3 mm {0.091 in}
2.2 mm {0.087 in}
2.1 mm {0.083 in}
2.0 mm {0.079 in}
1.9 mm {0.075 in}
1.8 mm {0.071 in}

11. Assemble the selected snap ring.

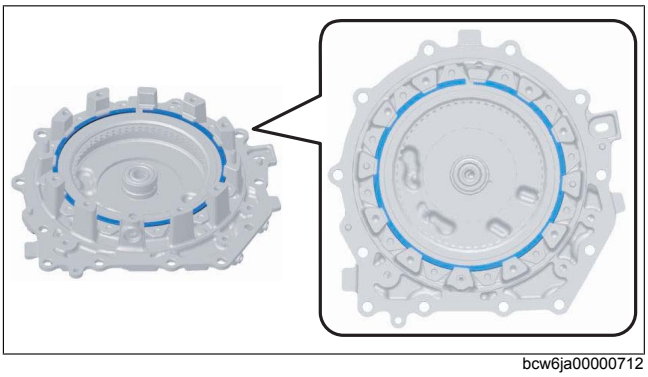
Caution
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

Note
- Snap ring size: Outer diameter approx. 174.2 mm {6.858 in}



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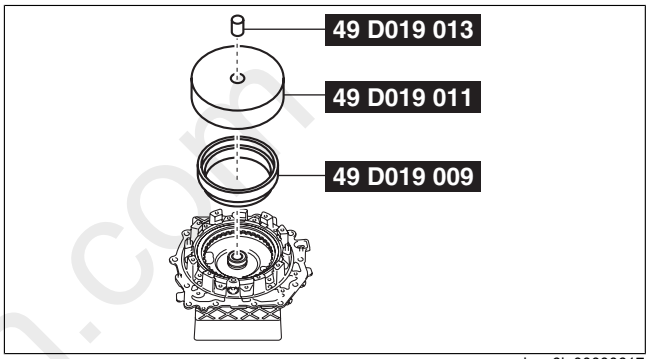


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12. Install the SSTs.

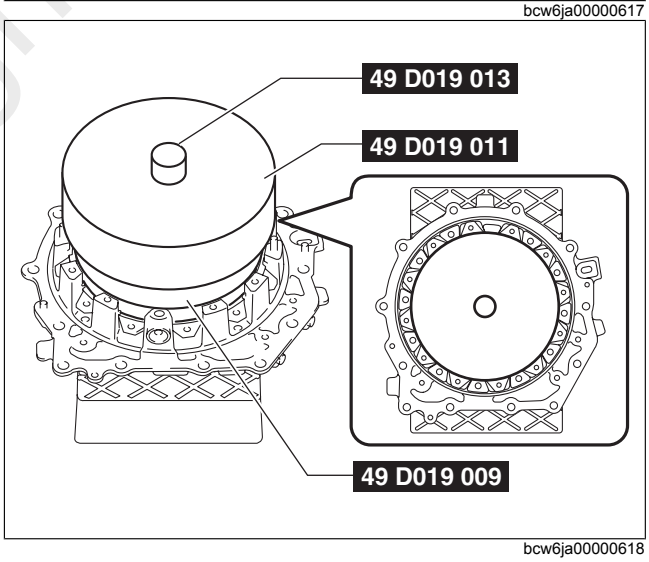
Caution

- **Install the SST (49 D019 009) so that the small diameter side is pointing towards the retaining plate.**



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13. Perform the R-3-5 brake clearance measurement from Step 6 (2). (See R-3-5 Brake Clearance Measurement.)



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